

Automated Circuit Discovery for NIDS

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Beyond the Black Box: ACDC to Reveal the Causality of Network Intrusions

Presented by: **Anbuselvi S**

Presentation Outline

- 1 Motivation & Background
- 2 Technical Framework
- 3 Research Applications
- 4 SOC-Oriented Integration
- 5 Conclusion

Motivation: The Interpretability Gap in Network Security

- Modern network security systems increasingly rely on **deep learning** models for automated threat detection.
- **The Black-Box Problem:** These models generate alerts, but their **internal decision logic** is opaque to human analysts.
- **Operational Impact:** Security teams face **high false-positive rates**.
- **Key Challenge:** Understanding *why* a model raises an alert, not just *that* it raises one.
- Motivation: **causal interpretability** > correlational explanations.

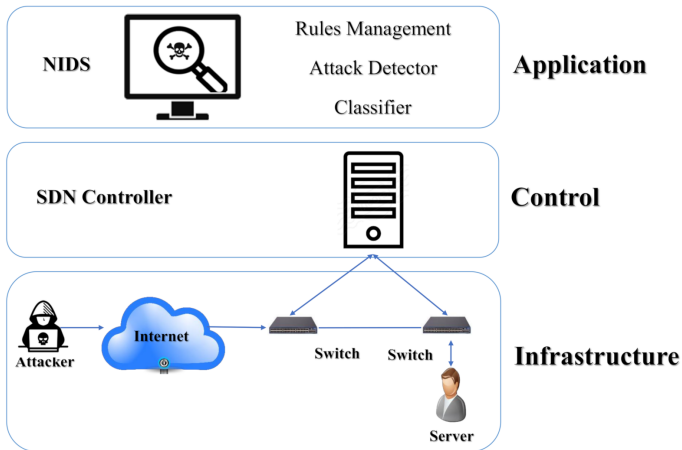
What is a Network Intrusion Detection System (NIDS)

- A **Network Intrusion Detection System (NIDS)** monitors network traffic to identify **malicious activities or intrusions**.
- It analyzes **packets, flows, and traffic patterns** traversing the network.
- NIDS raises alerts when observed traffic deviates from **normal or expected behavior**.
- It supports **Security Operations Centers (SOC)** in detecting attacks such as DoS, scanning, malware, and data exfiltration.

Key idea:

“Is this network traffic normal or malicious?”

Network Intrusion Detection System (NIDS)



High-level architecture of a Network Intrusion Detection System.

Interpretability Methods in NIDS: A Comparison

Aspect	Feature Attribution (SHAP / LIME)	Rule-Based XAI	ACDC (Proposed)
Explanation Level	Input features	Human-defined rules	Internal computational components
Causality	Correlational	Assumed by rules	Causal (via activation patching)
Model Internals	Ignored	Not applicable	Explicitly analyzed
Attack-Specific Logic	Weak	Manual	Automatically extracted
SOC Usefulness	Limited	High but rigid	High and operationally actionable
Forensic Value	Low	Moderate	High (circuit-level audit)

Applying ACDC to NIDS: Core Idea

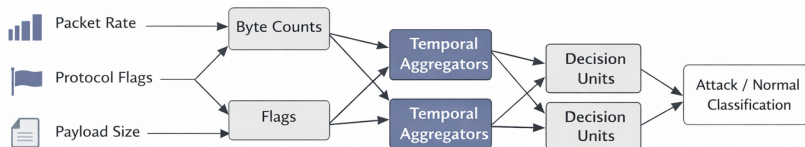
- In NIDS, the target behavior is an **attack detection decision** (e.g., SYN flood, scan, exfiltration).
- ACDC treats the trained NIDS as a **signal-processing system** that transforms raw traffic statistics into security decisions.
- The objective is to identify the **minimal internal computation** that maps:
 - traffic patterns (rates, flags, temporal bursts)
 - to an attack-specific alert.
- The extracted circuit reveals **which traffic characteristics** and **which internal interactions** the NIDS truly relies on.

Mapping ACDC Concepts to NIDS

ACDC Concept	Interpretation in NIDS
Behavior	Specific attack type (e.g., DDoS, Scan, Brute Force)
Computational Graph	NIDS model layers and feature interactions
Activation Patching	Replacing malicious traffic activations with benign baselines
Edge Pruning	Removing causally irrelevant feature pathways
Circuit	Minimal computation responsible for attack detection
Faithfulness Metric	KL divergence between full and pruned model outputs

System Architecture: ACDC-Enabled NIDS

NIDS as a Computational Graph



Step A: Task and Dataset Setup

To find a circuit, we must define the behavior:

Task Definition

Detecting SYN Flood attacks within the UNSW-NB15 dataset.

- **Clean Dataset:** Malicious traffic samples.
- **Corrupted Dataset:** Normal baseline traffic or mean-ablated features.

Behavior is quantified using the logit difference between Attack and Normal classes.

Datasets for Attack-Specific Circuit Discovery

(a) Basic statistics of datasets

Dataset	Labels	Features	Samples
CICIDS-2017	7	78	2.78M
RoEduNet-SIMARGL2021	3	29	31.43M
NSL-KDD	5	41	148K

(b) Distribution of samples across attack types

Dataset	Normal	DoS	PortScan	Brute	Web	Bot	Probe	U2R
CICIDS-2017	84.4%	9.1%	5.7%	0.5%	0.16%	0.07%	—	—
RoEduNet-SIMARGL2021	62.2%	24.5%	13.3%	—	—	—	—	—
NSL-KDD	48.0%	35.0%	—	—	—	—	10.1%	0.47%

These datasets enable attack-specific behavior isolation and evaluation of circuit faithfulness under realistic class imbalance.

Step B: Pruning Criterion (τ)

ACDC removes an internal connection ($w \rightarrow v$) if it does not causally influence the attack detection outcome.

$$D_{KL}(G \parallel H_{\text{new}}) - D_{KL}(G \parallel H) < \tau$$

- G : Full NIDS model predicting attack vs. normal traffic.
- H : Current candidate circuit under evaluation.
- D_{KL} : Change in output probability distribution.

KL divergence ensures the circuit preserves the **decision logic**, not merely classification accuracy.

Step C: Activation Patching

- Internal activations from **corrupted (benign) traffic** are injected into the forward pass of **malicious traffic**.
- If the attack prediction remains unchanged, the patched connection is **not causally required** for detection.
- This allows ACDC to distinguish **true attack logic** from incidental correlations.
- **Example:** replacing SYN packet-rate activations with statistics from normal background traffic.

Research: Generalization vs. Spurious Correlation

- Does the train NIDS learn **attack-relevant traffic patterns** (e.g., TCP flags, packet-rate bursts, temporal structure)?
- Or does it rely on **incidental identifiers** (e.g., source IPs, ports, or dataset-specific artifacts)?
- ACDC reveals whether the extracted circuit encodes **robust attack semantics** or shortcut-based correlations.
- This analysis helps distinguish **true generalization** from spurious detection behavior.

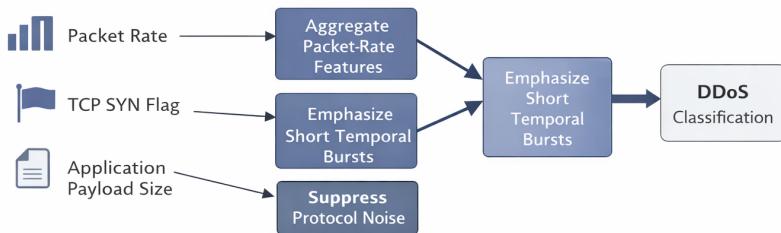
Example: Circuit Discovery for DDoS Detection

- Consider a trained NIDS tasked with detecting **volumetric DDoS attacks**.
- ACDC isolates a **minimal detection circuit** that:
 - aggregates packet-rate signals across short time windows,
 - emphasizes sudden temporal traffic bursts,
 - suppresses protocol-specific and benign background noise.
- This circuit constitutes the **causal computation** sufficient for triggering a DDoS alert.

ACDC-Based Circuit Extraction for DDoS Attacks

Example: Circuit Discovery for DDoS Detection

ACDC isolates a DDoS-specific circuit within the trained NIDS



Technical Challenge: Scalability

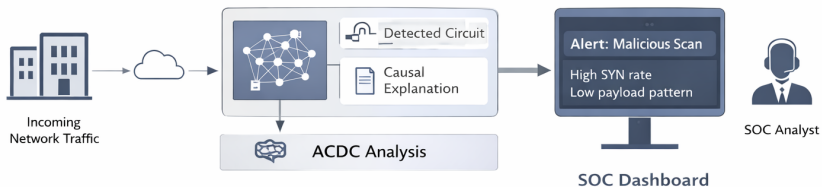
- Activation patching is computationally expensive for high-throughput network traffic.
- **Research Focus:** Attribution patching to approximate circuit importance with significantly reduced computational cost.
- As a result, ACDC is currently suited for **offline analysis and post-incident investigation**, rather than inline detection.

Failure Case: The OR-Gate Problem

- **Redundancy:** A NIDS may learn multiple internal pathways that independently detect the same attack.
- **Pruning Risk:** If removing one pathway does not affect performance, ACDC may prune both during causal testing.
- **Interpretation:** This reflects redundancy in detection logic—not model failure.
- **Research Challenge:** Handling redundant causal pathways remains an open problem in circuit discovery.

SOC Integration: System Design

SOC-oriented Explainable Alerts



Traditional Alert: *Anomaly detected*

ACDC-Enhanced Alert

Logic: Observed high SYN count combined with low data payload patterns.

Causal confidence: 99.2%.

Circuits are summarized, not shown raw, to analysts.

- **Causal Visibility:** For every "Transparent Alert," ACDC allows analysts to expand the view to see the **internal detection circuit**.
- **Behavioral Logic:** Analysts can inspect exactly which traffic patterns (e.g., specific packet headers or temporal spikes) interacted to trigger the model's "high-risk" state.
- **Incident Attribution:** Supports faster identification of **Zero-Day threats** by identifying novel circuit structures that don't match known attack signatures.

Circuit Level Debugging for NIDS Models

- **The Bottleneck:** Traditional NIDS are "Black Boxes"—fixing one error often requires retraining the entire model.
- **ACDC-Enabled Debugging:**
 - **Pinpoint:** Identify the specific sub-circuit responsible for a false alert.
 - **Causal Validation:** Confirm the error using activation patching with "clean" baseline traffic.
 - **Targeted Update:** Refine only the implicated nodes, leaving the rest of the model's logic intact.
- **Operational Value:**
 - Faster time-to-patch for SOC teams.
 - Eliminates the risk of "catastrophic forgetting" during global retraining.

Evaluation: Quantifying Model Trust

- **How do we know the refined circuit is reliable?**
- **Faithfulness (D_{KL}):** Does the simplified circuit still "think" like the full model?
- **Robustness (Accuracy):** Does detection remain high after we prune the "noise"?
- **Interpretability (Sparsity):** Is the circuit small enough for a human to audit in under 60 seconds?

Step 1: Faithfulness & Complexity

- **KL Divergence (D_{KL}):** Measures the informational "loss" when moving from the full model to the circuit.
 - *Goal:* $D_{KL} \approx 0$ indicates the circuit logic is identical to the model.
- **Logit Margin Analysis:** Measures how much the model's "confidence" drops when specific edges are ablated.
- **Circuit Sparsity (Efficiency):**
 - Focuses on the "Occam's Razor" of security: The simplest explanation is usually the most auditable.
 - Metric: Percentage of edges pruned while maintaining accuracy.

Step 2: Operational Validation in NIDS

- **Recovery Quality:** Proof that the circuit captures the *actual* attack signatures, not just random noise.
- **Corrupted Traffic Stress Test:**
 - Does the circuit break if we corrupt a malicious feature?
 - If yes, the circuit has successfully identified a **causal feature**.
- **Q1/Q2 Logical Integrity:**
 - **Q1 (Sufficiency):** Is the circuit enough to detect the attack?
 - **Q2 (Minimality):** Does removing any more edges break the detection?

Building Operational Trust

- **Auditability:** ACDC provides a causal explanation for each intrusion detection decision.
- **Accountability:** Enables post-incident review and justification in regulated and security-critical environments.
- **Inspectability:** Improves analyst confidence by making AI-driven decisions transparent rather than opaque.

- Framed network intrusion detection as a **mechanistic interpretability problem**, rather than a black-box prediction task.
- Applied Automatic Circuit Discovery (ACDC) to extract **causal, attack-specific computational circuits** from trained NIDS models.
- Demonstrated that circuit-level explanations enable **auditable and trust-oriented decision support** for SOC operations.

Future Directions: The Roadmap to Trusted NIDS

- **Vision:** Moving from black-box detection to an auditable, interactive security ecosystem.
- **Real-Time Scalability**
Approximate ACDC and attribution patching for high-throughput, active network environments.
- **Advanced Threat Discovery**
Zero-day behavioral analysis and resolving "OR-Gate" redundancy in overlapping attack logic.
- **The Human-in-the-Loop SOC**
Automated circuit-to-rule translation and interactive tools for AI-driven decision refinement.

Thank You!